Mohammad Khalid

Frontend Engineer | React, Next.js, TypeScript, Tailwind CSS, WebGL, Web Performance & SEO
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Right to work in the UK until 30 June 2028

Summary

Frontend-focused software engineer (BSc Information Technology, 2024) who builds high-performance, well-tested web and mobile applications with rich interaction and fluid animation. Creator of Kepler (open-source Android audio & e-reader app in Jetpack Compose) and Bustan Bakers (production Next.js site with CI-enforced performance budgets). Experienced with three.js, WebGL2/GLSL shaders, Svelte 5, and state machines. Uses AI coding agents daily with strict TypeScript, automated testing, and CI/CD pipelines. Currently IT Support Specialist at S&P Global, London.

Technical Skills

- **Frontend:** React 19, Next.js (App Router, SSR/SSG), TypeScript, JavaScript (ES6+), Svelte 5, HTML5, CSS3, Tailwind CSS, Framer Motion, three.js (WebGL2, GLSL), CSS scroll-driven animations, responsive UI, accessibility (WCAG)
- **Mobile UI:** Kotlin, Jetpack Compose, Material 3 Expressive, Readium EPUB engine, Flutter, Dart, StateFlow
- **Performance & SEO:** Core Web Vitals (LCP, INP, CLS), bundle size budgets, AVIF & responsive images, code splitting, structured data (JSON-LD), Open Graph, sitemaps
- **Testing & CI/CD:** GitHub Actions CI/CD, Node.js test runner, unit & integration tests, ESLint, Vercel, GitHub Pages, Git
- **Backend & APIs:** Node.js, RESTful APIs, OAuth 2.0 (GitHub), WebSocket streaming, Room 3 (SQLite), Firebase (Auth, Firestore)
- **AI & Tooling:** Claude Code, OpenAI API, Google Gemini API, OpenRouter API
- **Other Languages:** Python, C (C99), Bash

Projects

Kepler (github.com/iamkhalid2/Kepler) | Kotlin, Jetpack Compose, Readium, Media3 | 2026 - Present

- Built an open-source Android reader and audio narration app using Jetpack Compose and Material 3 Expressive, featuring a custom Readium EPUB rendering engine and responsive gesture navigation.
- Integrated AndroidX Media3 for background playback with lockscreen media controls, seamless chapter navigation, and reactive UI state management via Kotlin Coroutines and StateFlow.

Portfolio (iamkhalid.xyz) | Next.js 16, React 19, TypeScript, three.js, Tailwind CSS | 2024 - Present

- Built a scroll-driven homepage as a single three.js WebGL2 renderer with raw GLSL shaders (Schwarzschild geodesics, Doppler and gravitational colour shifts, HDR bloom), keeping per-frame state outside React.
- Rendered portrait with ~28,000 particles streaming on scroll; implemented pinned scenes, Lenis smooth scroll, custom cursor, command palette, and a git-backed CMS with GitHub OAuth.

Bustan Bakers (bustanbakers.com) | Next.js, React, TypeScript, Tailwind CSS, WebGL | 2025 - Present

- Built and deployed a statically generated multi-page site on Vercel with scroll-driven CSS animation, a custom cursor, a handwritten-wordmark splash with a morphing page transition and a lazy-loaded WebGL hover shader.
- Set up a GitHub Actions pipeline with strict performance budgets (178 KB gzipped JS limit), automated per-route SEO checks (canonical URL, JSON-LD, Open Graph), and an AVIF image pipeline with five responsive widths.

Invent a GPT course (iamkhalid2.github.io/microgpt-course) | Svelte 5, Vite, Python, Pyodide | 2026 - Present

- Built a 19-chapter interactive web course in Svelte 5 and Vite where learners execute Python in-browser via Pyodide, with real-time transformer architecture visualizations (12 GitHub stars).
- Deployed to GitHub Pages with GitHub Actions; verified in-browser model outputs against Python reference implementations.

Also: Kiraya (Flutter, Firebase, 2025), a property management app with cloud sync. Computer-Use (Python, Gemini Live API, 2025), a voice-controlled desktop agent (11 stars). WhatsAppGPT (Node.js, 2023), a WhatsApp chatbot on OpenAI and Gemini (17 stars, 15 forks).

Experience

IT Support Specialist | S&P Global, London, UK | 2026 - Present

- Provide technical support to employees at a global financial intelligence and data company, handling incidents and service requests in ServiceNow.
- Support and troubleshoot enterprise devices and policy compliance with Microsoft Intune.
- Administer user accounts, security groups, and access controls in Active Directory.

Education

BSc Information Technology | Graphic Era University, Dehradun, India | 2021 - 2024